

World's smallest nanomotor that runs the longest

Engineers at the University of Texas at Austin, USA have developed world's smallest nanomotor which they claim to be the fastest-ever developed. The Nanomotor, as they call it, is 1/500th the size of a grain of salt! It is the fastest with ultra-high speed and is able to run the longest.

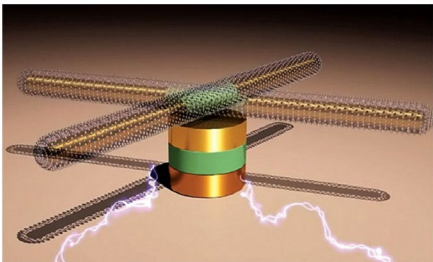
The Nanomotor is seen as a significant step towards manufacturing miniature machines. It can be used in first-of-its-kind devices, which can be fitted in applications that require unprecedented precision, such as controlled biochemical drug delivery to live cells and cell-to-cell communication.

One of the unique features of it is its large driving power. It is able to deliver 15 continuous hours of rotating speed at 18,000 rpm, equivalent to the speed of a motor in a jet airplane engine. The other nanomotors available today are able to run at speeds up to just 500 rpm and can last only for a few minutes.

The Nanomotor is capable of reaching and staying in places which were beyond imagination earlier. Courtesy its size of one micrometer, it can easily fit inside a human cell. One amazing aspect of the nanomotor is its field of nanoelectromechanical systems (NEMS). With this, the machine is able to push forward the frontiers of cheaper and more energy-efficient systems.

Coming soon: holographic projections from cell phones

Holographic projections from your very own smartphone could be a reality soon, suggests a new report by the *Wall Street Journal*. California-based start-up Ostendo Technologies has reportedly built the world's first-ever hologram projector chip that could emit videos and 3D images from smartphones and other screens.



World's smallest and fastest Nanomotor (Courtesy: www.crazyengineers.com)

A chipset when fitted into a smartphone will help project video on a surface with 122cm (48-inch) diagonal view. Several of these chipsets together will be able to beam larger, more complex images. "The first iteration of the chip will only project 2D videos but the next version, expected to follow soon after, will feature holographic capability," Hussein S. El-Ghoroury, founder of Ostendo was quoted as saying.

The chip essentially consists of an image processor with a wafer made up of miniaturised light-emitting diodes (LEDs). The system so formed can control the colour, brightness and angle of over a million individual beams of light, says the report.

These batteries can charge your phone in ten minutes

Researchers at the University of California, Riverside Bourns College of Engineering simply replaced standard graphite-based anodes in Li-Ion batteries with silicon, and the results were simply outstanding. Li-Ion batteries so obtained can charge a device like your power-hungry smartphone in just ten minutes!

Researchers Mihri and Cengiz Ozkan observed that, silicon with its

10x bigger charge capacity compared to regular graphite-based anodes allowed 63 per cent increase in the total cell capacity and makes the battery about 40 per cent lighter at the same time. Based on their observations, they built prototype batteries (using silicon as anode) that could be charged 16x faster compared to standard Li-Ion batteries.

Earlier, an Israeli startup called StoreDot came out with a prototype charger that can replenish your smartphone battery in half a minute. StoreDot recently demonstrated their prototype charger at Microsoft's Think Next Conference successfully charging a Samsung Galaxy S4's battery (2600mAh) in under half a minute. The catch is the charger works only with the Galaxy S4 for now.

Laser tech that can sense objects from ten metres

A new laser technology has been developed which can sense objects from as far as ten metres (approx. 30 feet). The technology, developed by researchers from the University of California, Berkeley can sense objects from 10 times farther than any other